

Manipulating Interfacial Water Configuration via Constructing Asymmetric Structure Unit for Hydrogen Production in Alkaline Seawater

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The configuration of interfacial water molecules plays a crucial role in affecting the alkaline hydrogen evolution reaction (HER). Here, a strategy of altering the water molecular configuration through element doping is proposed to enhance HER activity in seawater splitting. By doping the Mo—O group into FeP to construct a Mo—O—FeP asymmetric structure, *ab initio* molecular dynamics (AIMD) and *in situ* Raman results demonstrate that Mo—O doping changes the molecular configuration, and more free water on the catalyst surface is dissociated to continuously supply sufficient protons during the HER process, accelerating the HER kinetics and mass transfer process. Density functional theory (DFT) calculations reveal that the Mo—O—FeP catalyst achieves a more balanced H adsorption/desorption. Additionally, asymmetric structure reduces the adsorption energy of Cl[−] at the Fe site, preventing Cl[−] poisoning in seawater. Benefiting from the above, Mo—O—FeP can reach 1000 mA cm^{−2} with an overpotential of 279 mV in alkaline seawater and has excellent stability. In addition, an AEM electrolyzer assembled from Mo—O—FeP demonstrates outstanding activity and can maintain stable operation for more than 100 h.

The introduction of an alkaline substance in seawater can remove most of the Ca²⁺ and Mg²⁺ precipitates, thus reducing the toxic effect on the electrode. In alkaline seawater electrolyte, the dissociation of water molecules is a key step in the hydrogen evolution reaction (HER), which is often affected by the adsorption configuration of water molecules on the catalyst surface.^[3,4] As a polar molecule, water molecules will be polarized under the action of an electric field, which will lead to a change of molecular orientation and thus change the order degree of the interface network.^[5,6] The H-down configuration is considered the most favorable for water molecular dissociation. In the H-down configuration, the hydrogen atoms of the water molecules are oriented toward the surface of the catalyst, while the oxygen atoms are oriented away from the surface. This specific orientation can significantly shorten the distance between the active site (M) and the hydrogen atom, thereby

enhancing the binding energy of the M—H bond. This enhanced binding energy not only facilitates water dissociation but also promotes the subsequent H adsorption and H₂ formation process.^[7,8] For example, Liu et al.^[9] successfully achieved the redirection of interfacial water molecules through the induction of the local electric field of atoms, which significantly increased the proportion of H-down configuration. This configuration improved the dissociation efficiency of water molecules, thereby reducing the overpotential of HER and improving the overall

1. Introduction

Hydrogen energy is regarded as one of the most potential clean energy in the 21st century, and water electrolysis is an important pathway for clean hydrogen production. The seawater electrolysis technology can directly use seawater without a complex pretreatment process, thus reducing the cost and energy consumption of hydrogen production.^[1,2] Alkaline seawater electrolysis is a promising method for large-scale hydrogen production.

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electrocatalytic activity. In light of the aforementioned theoretical, it is imperative to develop appropriate catalysts to investigate the pivotal role of interfacial water for alkaline HER in seawater.

Transition metal phosphides is regarded as a potential catalyst for HER in electrolytic seawater due to its diverse chemical composition, unique structural characteristics, and excellent electrical conductivity.^[10,11] Nonetheless, in the complex seawater environment, the catalyst may be compromised by Cl^- and impurities, leading to structural instability or performance degradation, thereby affecting long-term stability and catalytic activity.^[12] Recent studies have demonstrated that the catalytic performance of phosphide electrocatalysts can be substantially enhanced by incorporating oxygen-philic substances (such as oxygen or hydroxyl groups).^[13–15] These oxyphilic substances can form hydrogen bonds with hydronium ions or water molecules at the electrode/solution interface, thereby fine-tuning the interfacial microenvironment. Therefore, this modulation may not only optimize the adsorption/desorption behavior of reaction intermediates but also facilitates the dissociation of water molecules.^[16,17] Consequently, the rational design and synthesis of M–O modified phosphide electrocatalysts to form asymmetric structures hold significant importance for enhancing HER kinetics in seawater.

Herein, we propose a strategy of designing Mo–O–FeP asymmetric structure to improve catalyst kinetics by changing the configuration of water molecules, so as to enhance HER performance in seawater electrolysis. In situ Raman spectroscopy and AIMD results show that the introduction of Mo–O forms an asymmetric structure, thus changing the configuration of water molecules, the dicoordinated water gradually transforms into free water, and more free water will continuously provide enough protons to the interface in the HER process, improving the H_2 production efficiency. In addition, the density functional theory (DFT) calculation shows that the center of the d band Mo–O–FeP is far away from the Fermi level, which is favorable for H^* desorption and subsequent H_2 generation. In addition, the Mo–O–FeP shows a weak Cl^- adsorption energy for seawater electrolysis. Benefiting from the above advantages, Mo–O–FeP needs an overpotential of only 279 mV to reach 1000 mA cm^{-2} in alkaline seawater and has excellent stability. Moreover, the AEM electrolyzer assembled by Mo–O–FeP shows excellent activity and stability over 100 h, with almost no activity attenuation in seawater electrolysis. This work emphasizes doping engineering to alter the configuration of water molecules to improve HER kinetics in industrial seawater electrolysis.

2. Results and Discussion

2.1. AIMD Calculations

In alkaline media, protons (H_3O^+) are nearly absent, with H_2O serving as the sole source of protons in alkaline electrolytes.^[18,19] This makes the dissociation of water a critical factor in the dynamics of the HER. To investigate the effect of asymmetric structural units formed by Mo–O introduction on interfacial hydrogen bonds of catalysts, we performed ab initio molecular dynamics (AIMD) simulations. The calculation models of Mo–O–FeP and FeP are shown in Figure S1 (Supporting Information). In both AIMD schemes, the solvation environment is consistent.

As shown in Figure 1a,b, the typical snapshots and the hydrogen bond distribution show that the average distance between water molecules and the surface of Mo–O–FeP decreases from 3.5 to 3.3 Å with the addition of Mo–O. It proved that the water molecules are closer to the catalyst surface after doping with Mo–O. This will enhance the interaction force between water molecules and the catalyst, which helps to improve the activation efficiency of water molecules, making it easier to dissociate and release active hydrogen species (such as H^+ or H^*), thereby promoting the progress of HER. When the distance between H_2O molecules and Mo–O–FeP exceeds 3.5 Å, there is no obvious structural difference between H_2O molecules on Mo–O–FeP and FeP surfaces compared to those in the bulk phase. However, when the distance falls below 3.5 Å, the number of hydrogen bonds on the surface of the Mo–O–FeP catalyst decreases, and the total number of hydrogen bonds is less than FeP (Figure S2, Supporting Information). The decrease in hydrogen bonds indicates that H_2O molecules at the catalyst interface are polarized, disrupting the ordering of the interfacial water network. thus, promoting the transfer of intermediates and making it easier for H_2O molecules to reach the catalyst surface (Figure 1c,d).^[20,21]

2.2. Preparation and Characterization of Mo–O–FeP

A molten salt and low-temperature phosphating strategy was used to prepare molybdenum oxygen doped ferric phosphide on foam iron (Mo–O–FeP). First, $\text{Fe}(\text{NO}_3)_3 \cdot 9\text{H}_2\text{O}$ is used as the main Fe source, and $(\text{NH}_4)_6\text{Mo}_7\text{O}_{24} \cdot 4\text{H}_2\text{O}$ is used as the Mo source to provide the MoO_4^{2-} ion. The two are treated by simple molten salt to form the MoFe component (MFC), and the Mo–O group is directly grown on the IF. Then, the MoFe complex components are phosphatized in an inert gas environment (such as N_2), and Fe^{3+} is reduced to Fe^0 to form FeP. Some Mo–O groups remain unchanged and exist as dopants in FeP. Finally, the Mo–O-doped FeP species in the form of nanoparticles are obtained. The morphology of the MFC precursor is characterized by scanning electron microscopy (SEM). Figure S3a (Supporting Information) shows the in situ growth of dense nanoarrays on the catalyst surface. Following the preparation of the MFC precursor, phosphating is performed to obtain the Mo–O–FeP catalyst. Both Mo–O–FeP and FeP exhibit similar nanoparticle morphologies, indicating that the morphology of catalysts doped with Mo do not change significantly (Figure S3b,c, Supporting Information). The rough nanoparticle surfaces tend to expose more active sites. The transmission electron microscopy (TEM) images of Mo–O–FeP shown in Figure 2a also confirm that the catalyst has different nanoparticle stacking structures. The high-resolution transmission electron microscopy (HRTEM) image (Figure 2b) clearly shows that the lattice spacing is 0.270 and 0.253 nm, corresponding to the (011) and (120) crystal planes of FeP, respectively. It is worth noting that the increased lattice distance of Mo–O–FeP with the (120) crystal plane may be attributed to element doping (Figure S4, Supporting Information). In addition, the typical TEM and corresponding elemental mapping show that Mo, Fe, O, and P are uniformly distributed throughout Mo–O–FeP (Figure 2c). And the corresponding EDS spectrum of Mo–O–FeP is shown in Figure S5 (Supporting

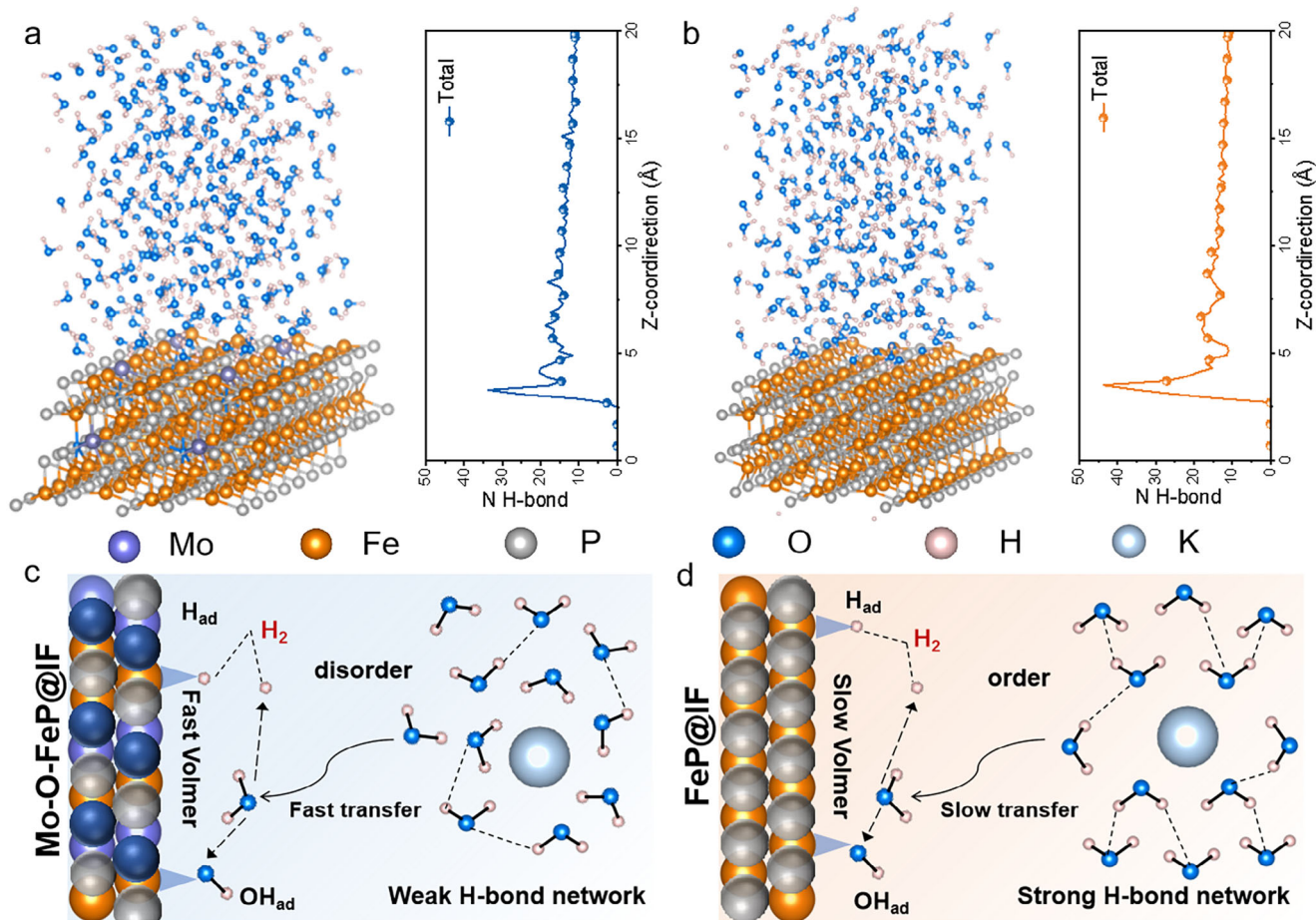


Figure 1. Representative snapshots of the H₂O structure and corresponding hydrogen bond distribution at the interface of (a) Mo–O–FeP and (b) FeP. Mechanism diagram of interaction between free water and (c) Mo–O–FeP and (d) FeP catalysts.

Information). The structure of the catalyst was characterized by X-ray diffraction (XRD). The diffraction peaks of Mo–O–FeP and FeP are well directed to FeP (JCPDS No.39–0809), which are consistent with the results of HRTEM, indicating that the addition of Mo–O does not alter the overall structure of the material. Notably, no distinct Mo peak is detected in the XRD pattern of Mo–O–FeP, which further confirms the doping of Mo (Figure S6a,b, Supporting Information). Furthermore, the Raman spectrum of Mo–O–FeP exhibits a prominent characteristic peak at 985 cm^{−1}, which is closely related to the stretching pattern of Mo–O bonds, confirming the successful introduction of [Mo–O] species into the FeP structure.^[22] The surface element composition and chemical state of the catalyst are analyzed by X-ray photoelectron spectroscopy (XPS). The full XPS spectrum of Mo–O–FeP indicates the presence of Mo, Fe, O, and P elements (Figure S7, Supporting Information). The XPS spectrum of Mo 3d confirms the successful doping of Mo. Specifically, the observed peak at 232.1 eV corresponds to Mo⁶⁺, while the peak at 235.2 eV is associated with the lower oxidation state of Mo⁵⁺ (Figure 2d).^[23] The binding energies of Fe 2p spectra at 728.4, 723.8, 714.7, and 710.6 eV correspond to Fe³⁺ 2p_{1/2}, Fe²⁺ 2p_{1/2}, Fe³⁺ 2p_{3/2}, and Fe²⁺ 2p_{3/2}, respectively.^[24] Additionally, there is also a peak belonging to the Fe–P bond at

706.7 eV, which further proves the successful synthesis of FeP substances. Notably, the binding energy of Fe 2p in Mo–O–FeP exhibits a positive shift compared to that in FeP (Figure S8a, Supporting Information). The P 2p spectrum of Mo–O–FeP exhibits three significant peaks. The peak at 133.2 eV is associated with the P–O bond, while the other two peaks correspond to the P 2p_{3/2} and P 2p_{1/2}.^[25] Notably, compared with FeP, the P 2p spectrum of Mo–O–FeP shows a relatively negative shift. The above results indicate that there is an electronic interaction between Mo–O and FeP (Figure S8b, Supporting Information). Furthermore, the XPS spectrum of O 1s can be deconvoluted into three peaks; the peak at the binding energy of 532.4 eV is attributed to P–O, and the remaining two peaks at 529.6 and 531.0 eV are attributed to metal oxide and O–H groups, respectively (Figure S8c, Supporting Information).^[26] The O 1s spectra of Mo–O–FeP display a relative negative shift compared to that of FeP, implying that electrons accumulate in the doped electronegative O atoms. The electron distribution between doped Mo–O and FeP is revealed by calculating the charge density difference of Mo–O–FeP. As shown in Figure 2e, the introduction of [Mo–O] species induces charge depletion around the Fe and Mo atoms (green regions) and promotes charge accumulation around the P and O atoms (yellow regions). Based on the

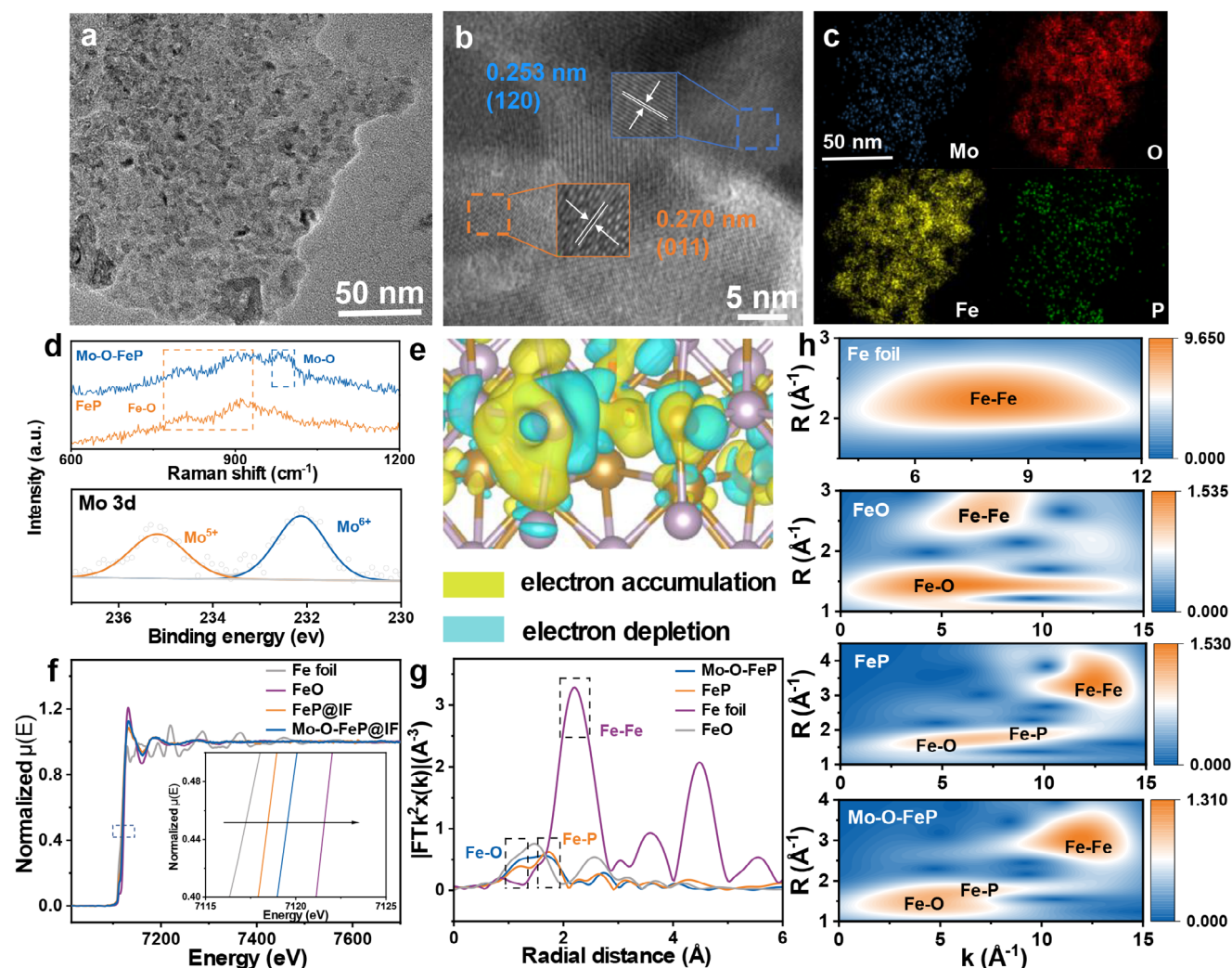


Figure 2. a) TEM and b) HRTEM of Mo—O—FeP. c) The corresponding element mapping images of Mo—O—FeP. d) Raman spectra and high-resolution XPS of Mo 3d of Mo—O—FeP. e) Charge density difference of Mo—O—FeP. f) Fe K-edge XANES and (g) EXAFS and h) the wavelet transforms plots of Fe foil, FeO, FeP, and Mo—O—FeP.

Bader charge calculation results, P and O atoms show electron gain, while Mo and Fe atoms show electron loss (Figure S9, Supporting Information). This result is consistent with the results of XPS and differential charge density analysis, further verifying the direction and extent of charge transfer. X-ray absorption near-edge structure (XANES) analysis shows that the average valence states of both Mo—O—FeP and FeP are within the 0–2 valence state. Notably, Mo—O—FeP exhibits a positive shift relative to FeP (Figure 2f), indicating that the electron density around Fe decreases after Mo—O doping, which is consistent with the results of XPS. The Fourier-transformed extended X-ray absorption fine structure (FT-EXAFS) spectrum data in Figure 2g reveal typical peaks representing Fe—O and Fe—P bonds have been observed at Mo—O—FeP and FeP, demonstrating successful synthesis of the FeP phase. Wavelet transform analysis of the EXAFS spectrum further reveals that Mo—O—FeP and FeP share the same coordination environment, consistent with the EXAFS findings (Figure 2h).

2.3. Electrocatalytic Performance of Mo—O—FeP for HER

In order to investigate the structural advantages of Mo—O—FeP, the electrochemical properties of the prepared Mo—O—FeP and the control samples are tested in 1.0 M KOH for HER. As shown in Figure S10a,b (Supporting Information), Mo—O—FeP demonstrates the lowest HER overpotential, requiring only 163 and 263 mV to achieve the current densities of 100 and 1000 mA cm^{−2}, which is significantly better than the control samples of MFC (253 and 423 mV) and FeP (153 and 323 mV). These results indicated that the introduction of Mo—O significantly increases HER activity. Tafel slope is a critical parameter to assessing the reaction kinetics of electrocatalysts for HER, derived from the LSV polarization curve. Compared with other samples, Mo—O—FeP has the lowest Tafel slope, indicating that the reaction kinetics of the Volmer-Heyrovsky mechanism is faster (Figure S11, Supporting Information). The charge transfer resistance between the catalyst and electrolyte is studied by electrochemical impedance

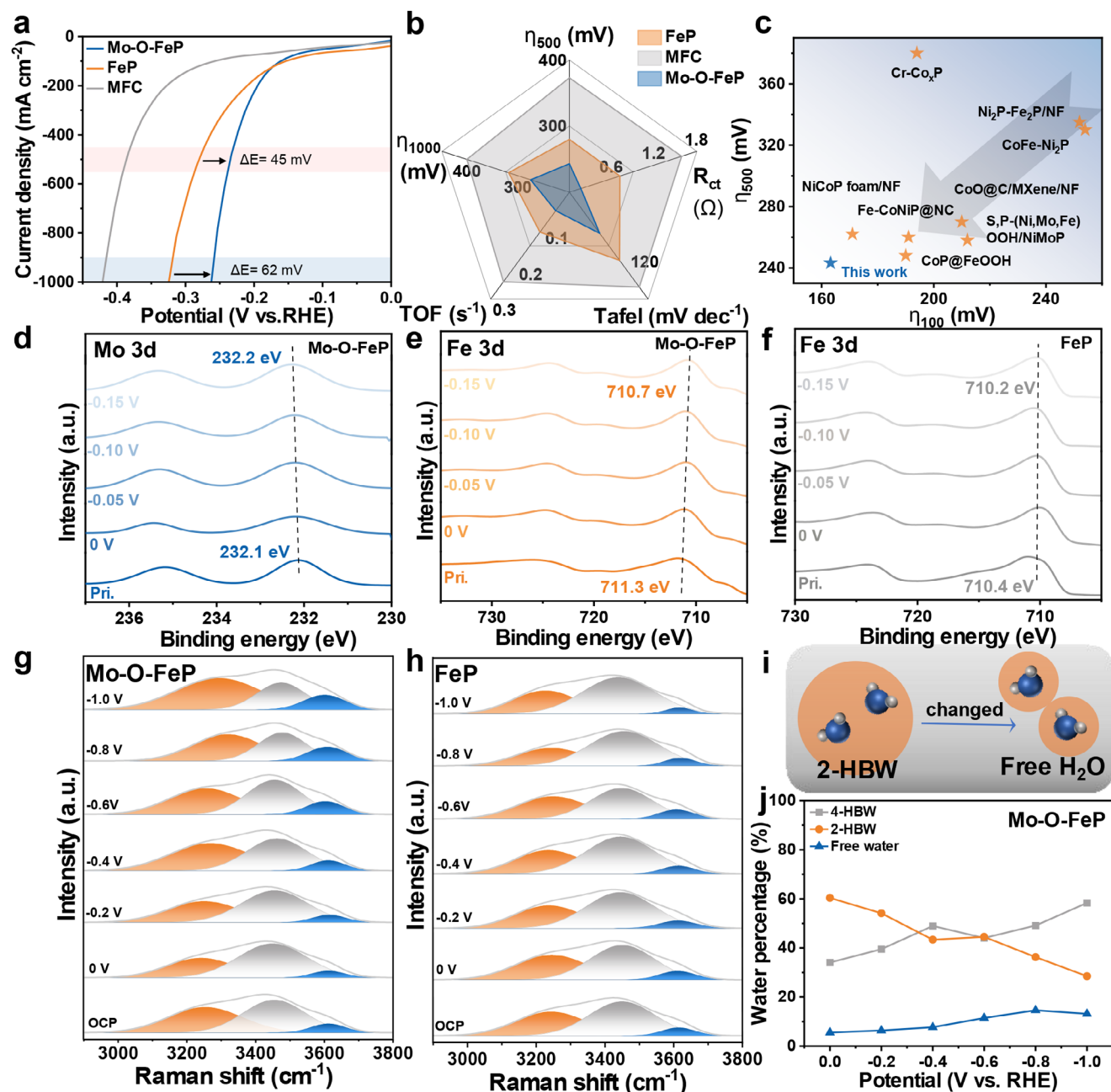


Figure 3. a) The LSV curves of Mo—O—FeP and control samples in alkaline seawater. b) HER performance comparison. c) Overpotential of Mo—O—FeP compared with other reported materials. d–f) Quasi in-situ XPS with Mo—O—FeP and FeP. g,h) In situ Raman of the studied catalysts. i) The proportion of different water molecular configurations for Mo—O—FeP at different voltages. j) Schematic diagram of structural transformation of water molecules.

spectroscopy (EIS). The Mo—O—FeP catalyst shows the smallest semicircle in all the samples (Figure S12, Supporting Information). This result suggests a rapid charge transfer rate between the electrode and electrolyte during electrocatalysis, further highlighting the efficiency of the Mo—O—FeP catalyst. Based on the excellent HER activity in 1.0 M KOH conditions, the electrocatalytic performance of Mo—O—FeP is further evaluated in the electrolyte of 1.0 M KOH + 0.5 M NaCl (simulated seawater) and 1.0 M KOH seawater (alkaline natural seawater). As shown in Figure 3a, in the 1.0 M KOH seawater, Mo—O—FeP only

needs an overpotential of 279 mV to reach the current density of 1000 mA cm⁻². This performance is comparable to the overpotential of 1.0 M KOH (263 mV) and 1.0 M KOH + 0.5 M NaCl (276 mV) (Figure S13, Supporting Information), indicating that Mo—O—FeP also exhibits potential for seawater splitting. Additionally, Mo—O—FeP exhibits the lowest Tafel slope and the smallest R_{ct} in alkaline natural seawater, further confirming its fast reaction kinetics and rapid charge transfer rate in this electrolyte (Figure S14, Supporting Information). Considering the effects of different load levels, the turnover frequency (TOF) indicates that

Mo—O—FeP has excellent intrinsic activity in alkaline seawater compared to other prepared catalysts (Figure S15, Supporting Information). To gain deeper insights into the effect of asymmetric structure formed by Mo—O doping on catalytic activity, we conduct temperature-dependent kinetic analysis for Mo—O—FeP and FeP to estimate the apparent activation energy (E_{app}) and pre-exponential factor (A_{app}) derived from Arrhenius equations for alkaline HER (Figure S16a–d, Supporting Information).^[27] Under the same overpotential, the E_{app} value of Mo—O—FeP is higher than that of FeP, indicating that the introduction of Mo—O groups enables Mo—O—FeP to have faster reaction kinetics, thereby improving HER activity. A_{app} generally reflects the number of active intermediates involved in the rate-determining step (Figure S17a,b, Supporting Information). Mo—O—FeP has a higher A_{app} at a lower E_{app} , resulting in better catalytic performance compared to FeP (Figure S17c, Supporting Information). Figure 3b shows a comprehensive comparison of HER performance among Mo—O—FeP, FeP, and MFC@IF, showing that the introduction of the [Mo—O] group significantly enhances HER performance even better than many recently reported non-precious catalysts in seawater (Figure 3c; Table S1, Supporting Information). Finally, durability is a crucial factor in evaluating the feasibility of scaling up hydrogen production from seawater. As shown in Figure S18 (Supporting Information), Mo—O—FeP can be stably electrolyzed for over 40 h in 1.0 M KOH seawater without significant current loss. Following the long-term durability test, it is found that the xps spectrum of P shows a peak of PO_4^{3-} . The surface of the phosphide may undergo partial oxidation under overpotential, generating species containing PO_4^{3-} . Phosphides have a strong tendency to oxidize under certain conditions, especially in high current density or a strong reducing environment. Even if it is a reduction potential, the local environment (such as high potential, strong reduction environment, etc.) and the chemical properties of the phosphide may promote the formation of PO_4^{3-} , or due to the fluctuation of electrode potential or the high-temperature effect of local hot spots, the surface phosphide may undergo partial oxidation to form PO_4^{3-} (Figure S19c, Supporting Information). This oxidation reaction is not directly triggered by external oxidants (such as oxygen), but rather due to local chemical changes on the electrode surface. Raman is a semi-quantitative means. As shown in Figure S20 (Supporting Information), the Raman peak of PO_4^{3-} appears when the potential reaches -0.4 V. When the voltage exceeds -0.4 V, the peak of PO_4^{3-} persists but does not increase significantly, further proving that the formation of PO_4^{3-} is promoted at a high reduction potential. And a small part of P is redeposited onto the surface of the catalyst. Furthermore, during the stability test, ICP-OES analysis is conducted on the solution at different time intervals to monitor the release of P during the operation of the catalyst (Figure S21, Supporting Information). The result shows that the dissolution rate of P element is relatively fast at the beginning of the stability test, and then gradually stabilizes. This change further proves that PO_4^{3-} is generated on the surface of the catalyst, effectively inhibiting the continuous dissolution of P. Some P elements dissolve, resulting in a decrease in the electron cloud density around the oxygen atoms. This increases the binding energy of the O1s peak and causes it to shift toward higher energy. However, the peak position of XPS of other elements remains unchanged, indicating that the valence states of these elements

are stable and the overall structure of the catalyst has not been damaged (Figure S19a–d, Supporting Information). SEM (Figure S22, Supporting Information) shows no obvious morphological changes, and the HRTEM (Figure S23, Supporting Information) also shows that no new chemical composition is formed, indicating that Mo—O—FeP has excellent structural stability and corrosion resistance. In order to further monitor the dynamic changes of the catalyst during the electrochemical reaction, we investigate the XPS spectra of Mo—O—FeP at different voltages. As shown in Figure 3d, with the increase of potential, the main peak of Mo 2p exhibits a slight shift toward higher binding energy, indicating that a small number of electrons are transferred from Mo to Fe during the electrochemical process. For Fe 2d, when the potential is increased to -0.15 V, the main peak of Fe moves 0.60 eV negatively that of pristine Mo—O—FeP, while the main peak of FeP moves only 0.2 eV negatively that of FeP (Figure 3e,f). The voltage increase due to Mo—O and FeP interaction can enhance the reducing ability of the electrode and increase the electron density around Fe.

To elucidate the internal and external factors contributing to the enhancement of Mo—O—FeP for HER, we use in situ Raman spectroscopy to investigate the effect of interfacial water on the catalyst. We collected in situ Raman spectra of both Mo—O—FeP and FeP at varying potential, as shown in Figure 3g,h. Obvious wide peaks can be observed in the range of 3000 – 3800 cm^{-1} , which is attributed to the stretching mode of H_2O molecules. We decomposed this wide peak into three Gaussian peaks, which belong to three types of O—H stretching vibration, respectively.^[28] The peak ≈ 3200 cm^{-1} belongs to tetrahedron-coordinated hydrogen-bonded water species (4-HBW), and the peak ≈ 3400 cm^{-1} is associated with dicordinated hydrogen-bonded water species (2-HBW). Additionally, the peak observed at ≈ 3600 cm^{-1} is attributed to the free water (KW).^[29,30] 4-HBW is a condensate of strongly hydrogen-bonded water.^[31–33] Studies have shown that a high content of strongly hydrogen-bonded water is a key factor for the relatively low energy barrier of additives in the proton-coupled electron transfer process of the hydrogen bond network in interfacial water. Therefore, strongly hydrogen-bonded water played an indispensable role in promoting the transfer of hydrogen through the water-electrode interface.^[32,34] The in situ Raman spectroscopy indicated that the proportion of 4-HBW increases with the increase of voltage. The increased 4-HBW can enhance the active sites of the reaction. This is because 4-HBW can effectively participate in the intermediate reaction of hydrogen generation and serve as a catalytic medium for the reaction. Water molecules interact with electrode materials to reduce the reaction energy barrier, which is conducive to the precipitation of hydrogen, thereby significantly enhancing the alkaline HER activity. It is worth noting that free water represents a unique structure of interfacial water in alkaline media, and has an H-down structure. At a bias voltage of 0 – 0.1 V relative to reversible hydrogen electrode (RHE), Mo—O—FeP with the higher negative potential leads to more free water, and the proportion of free water increases from 5.5% to $\approx 14\%$. In contrast, the amount of 2-HBW shows a significant decline, from 60.3% to 28.5%. This trend indicates that the asymmetric structural units formed by the introduction of Mo—O alters the configuration of water molecules, and the 2-HBW gradually transforms into free water as the voltage increases (Figure 3i). More free

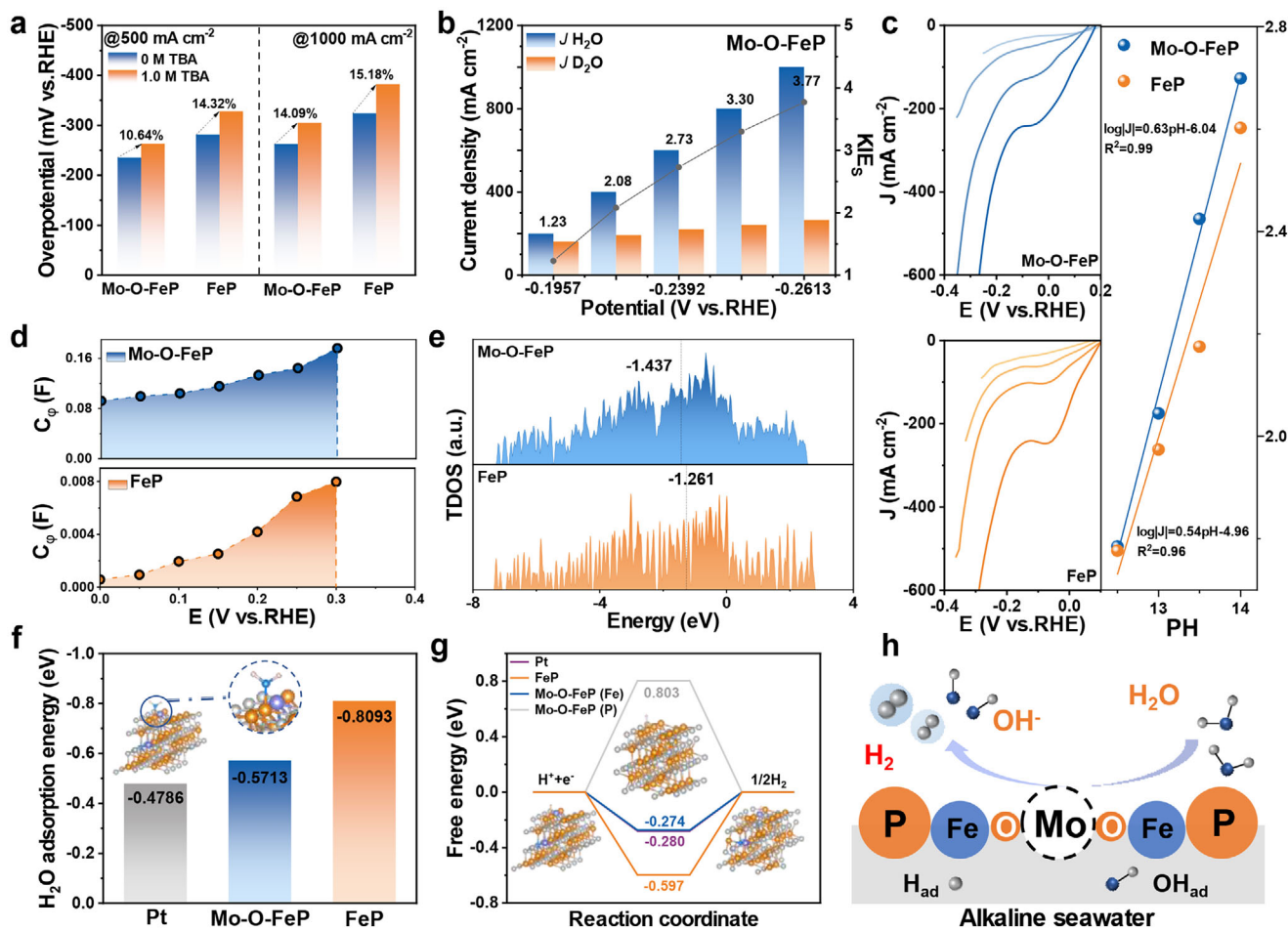


Figure 4. a) The KIE values under corresponding potentials of Mo—O—FeP. b) Overpotential comparison of Mo—O—FeP and FeP with different TBA concentrations. c) The diagram of pH-dependent. d) Plots of C_{dl} versus overpotential of Mo—O—FeP and FeP during HER in alkaline seawater solution. e) The TDOS of Mo—O—FeP and FeP. f) The H_2O adsorption energy and g) the ΔG_{H^*} of Mo—O—FeP, FeP, and Pt. h) HER mechanism of Mo—O—FeP.

water will continuously provide enough protons to the interface during the HER process, improving the electron transfer efficiency, thus accelerating the Volmer step of HER. In comparison, the proportions of various water configurations of FeP maintain a relatively stable trend (Figure 3j; Figure S24, Supporting Information). Therefore, it can be concluded that the asymmetric structure formed after the introduction of Mo—O changes the configuration of interfacial water molecules and accelerates the adsorption and dissociation of H_2O on the surface of Mo—O—FeP, which is consistent with the calculated results of AIMD.

2.4. Experimental Evidence for Enhanced H^* Adsorption of Mo—O—FeP

In order to investigate the impact of changes in interfacial water configuration on the kinetics of Mo—O—FeP in alkaline HER electrocatalysis, we conducted a series of electrochemical experiments and comprehensive tests. To assess the kinetics of H^* adsorption on FeP, we first use tert-butyl alcohol (TBA) as a specific hydrogen radical quencher to study the hydrogen adsorption

strength (Figure 4a; Figure S25, Supporting Information).^[35,36] Upon the addition of 1.0 M TBA, the HER activities of both Mo—O—FeP and FeP are significantly reduced. Specifically, at current densities of 500 and 1000 $mA\ cm^{-2}$, the potential of Mo—O—FeP increases by 10.64 and 14.09%, respectively, which is lower than the increases observed of FeP (14.32%, 15.18%) after adding 1.0 M TBA. This suggests that Mo—O—FeP is less affected by TBA, indicating a relatively stronger hydrogen adsorption capability compared to FeP. Additionally, the hydrogen/deuterium (H/D) kinetic isotope effect (KIE) can not only provide insights into the kinetics of hydrogen transfer but also reflect the water dissociation process. When the KIE value exceeds 1.5, it indicates that the water dissociation process is the rate-determining step of HER.^[37,38] Obviously, the HER activity of the catalyst all decreases in 1.0 M KOD + D_2O (Figure 4b; Figure S26, Supporting Information), but the KIE value of Mo—O—FeP is lower than that of FeP, which indicates that Mo—O doping promotes the water dissociation process and thus enhances HER activity. We further investigate the pH-dependent HER kinetics in alkaline media, and observe that HER performance increases with increasing pH (Figure 4c). Notably, Mo—O—FeP has a larger reaction order and a more pronounced pH dependence, indicating a more

efficient proton transfer capability.^[39] Moreover, we conducted in situ impedance tests to assess the strength of H⁺ adsorption on the catalyst (Figure S27 and Table S2, Supporting Information). By fitting the field impedance spectrum, we determine the capacitance ($C\phi$), which reveals that the integral area for Mo—O—FeP is greater than that for FeP, indicating that the hydrogen adsorption capacity of Mo—O—FeP is stronger (Figure 4d).^[40] In addition, the lower slope ($\log R_{ct}$ vs overpotential) further indicates that the incorporation of Mo—O accelerates the H⁺ adsorption kinetics (Figure S28, Supporting Information). And the peak of phase angle in the Bode diagram decreases rapidly, indicating that the surface of the Mo—O—FeP catalyst is conducive to the adsorption of reaction intermediates in the HER process (Figure S29, Supporting Information). Density functional theory (DFT) is adopted to further understand the mechanism of Mo—O—FeP catalyst enhancing HER activity in seawater, and the inherent catalytic activity is further understood by calculating the density of states (DOS) (Figure 4e; Figure S30, Supporting Information). Compared with FeP (−1.261 eV), the d band center of Mo—O—FeP is −1.437 eV. The d-band center far from the Fermi level is beneficial for subsequent H₂ generation and promotes the HER. We also compared the water adsorption energies on Mo—O—FeP and FeP (Figure 4f), with the corresponding adsorption configurations are shown in Figure S31 (Supporting Information). Mo—O—FeP shows a higher water adsorption energy of −0.8093 eV, which is more favorable for water adsorption than FeP (−0.5713 eV), so as to promote the subsequent water dissociation process. The free energy of hydrogen adsorption (ΔG_{H^*}) is a critical indicator of thermodynamic behavior and the catalytic enhancement mechanism of the catalyst. First, by comparing the adsorption energies of the Fe site and the P site, we determined that the adsorption energy of the Fe site is significantly higher than that of the P site. This suggests that the Fe site exhibits a stronger adsorption capacity for hydrogen and serves as the primary active site. And compared with FeP (−0.597 eV), the ΔG_{H^*} value of Mo—O—FeP is −0.274 eV, which is even better than the adsorption energy of Pt, indicating that Mo—O—FeP exhibits a stronger kinetic for HER in alkaline seawater (Figure 4g).^[41] In summary, the asymmetric structure formed by the introduction of Mo—O significantly promotes the kinetics of alkaline HER and accelerates H₂ production by changing the configuration of interfacial water (Figure 4h).

2.5. Evaluation of Corrosion Resistance of Mo—O—FeP in Seawater and Application of AEM Electrolyzer

During the electrolysis of seawater, chloride ions may adsorb onto the active sites, leading to catalyst poisoning. Corrosion-resistant metal doping is an effective strategy against Cl[−] corrosion. Introducing the corrosion-resistant metal Mo into FeP not only enhances its catalytic activity but also improves its resistance to Cl[−] corrosion at high salt concentrations, thereby promoting efficient seawater cracking. Theoretical analysis indicates that the doping of Mo—O can alter the electronic structure of the FeP surface, promote the strong adsorption of hydrogen by the FeP surface, and simultaneously weaken the adsorption of Cl[−]. So, we further investigate the ability of Mo—O doping to exclude Cl[−] in

seawater. Figure 5a shows the adsorption energy of Cl[−] at the Fe site on the surfaces of FeP and Mo—O—FeP. Figure S32 (Supporting Information) is the Cl[−] adsorption model of FeP. Compared with pure FeP (−2.46 eV), the adsorption energy of Cl[−] at the Fe site of Mo—O—FeP is higher (−1.94 eV), so the ability of Mo—O—FeP to inhibit Cl[−] adsorption is stronger, which can effectively avoid the catalyst being poisoned by Cl[−]. In addition, as shown in Figure S33 (Supporting Information), the corrosion polarization curve data reveal that both the corrosion potential (E_{corr}) and corrosion current (I_{corr}) of the Mo—O—FeP electrode increase compared with FeP, which further proves that the catalyst doped with Mo—O is not susceptible to Cl[−] toxicity in seawater. As the process of electrolysis progresses, the concentration of Cl[−] in seawater can continue to increase. Therefore, we also assessed the HER properties of FeP and Mo—O—FeP in electrolytes with varying salt concentrations (Figure 5b,c).^[42] The Mo—O—FeP catalyst demonstrates excellent salt resistance, with the increase of salt concentration, the performance remains stable and even improves in 1.0 M and 5.0 M NaCl solutions. On the contrary, the FeP catalyst without Mo—O doping is easily poisoned by Cl[−], resulting in a decline in activity (Figure 5d). The above experimental results all confirm that Mo—O doping enhances the catalyst repulsion ability toward Cl[−]. The corrosion resistance of Mo—O—FeP in seawater has been further validated through the assembly of an alkaline seawater anion exchange membrane (AEM) electrolyzer, utilizing Mo—O—FeP as the cathode and S-(Ni, Fe)OOH (S-N, F) as the anode for seawater electrolysis (Figure 5e).^[43] For comparison, we also use commercial Pt/C as the cathode. As shown in Figure 5f, this battery exhibits an ultra-low voltage of 1.87 V to achieve a high current density of 1 A cm^{−2}, superior to commercial Pt/C@IF||S-N, F. In addition, we calculate the electrolysis efficiency of Mo—O—FeP||S-N, F, which reach 66.82% at a current density of 1000 mA cm^{−2}. This efficiency is a crucial indicator of the economic viability of electrolyzers, with the cost per GGE of H₂ is as low as \$1.003, about half of the US Department of Energy's target of \$2.00 by 2026, as detailed in Experimental Section (Figure 5g). The durability of the Mo—O—FeP||S-N, F electrolytic cell demonstrates that it could run at 100 mA cm^{−2} for more than 100 h without a significant increase in voltage (Figure 5h). The catalyst in the AEM electrolytic cell after 100 h of seawater electrolysis is characterized using SEM. The results demonstrated that the catalyst retained its original nanoparticle morphology, and the electrode structure did not change significantly. This finding confirms that the catalyst possesses excellent structural stability during prolonged operation (Figure S34, Supporting Information). Furthermore, during the stability test, ICP-OES analyses are performed on the solutions at various time intervals to monitor the potential release of metal ions during the operation of the catalyst, thereby providing additional evidence for the catalyst stability (Figure S35, Supporting Information). The result shows that the dissolution rate of metal ions is relatively high in the first 10 h and then gradually stabilizes, further proving the structural stability. The illustration in Figure 5h indicates that although the load capacity of Mo—O—FeP||S-N,F is higher than that of Pt/C@IF, the mass activity of Mo—O—FeP||S-N, F is higher. This finding further confirms the excellent catalytic activity of Mo—O—FeP||S-N, F in alkaline seawater electrolysis.

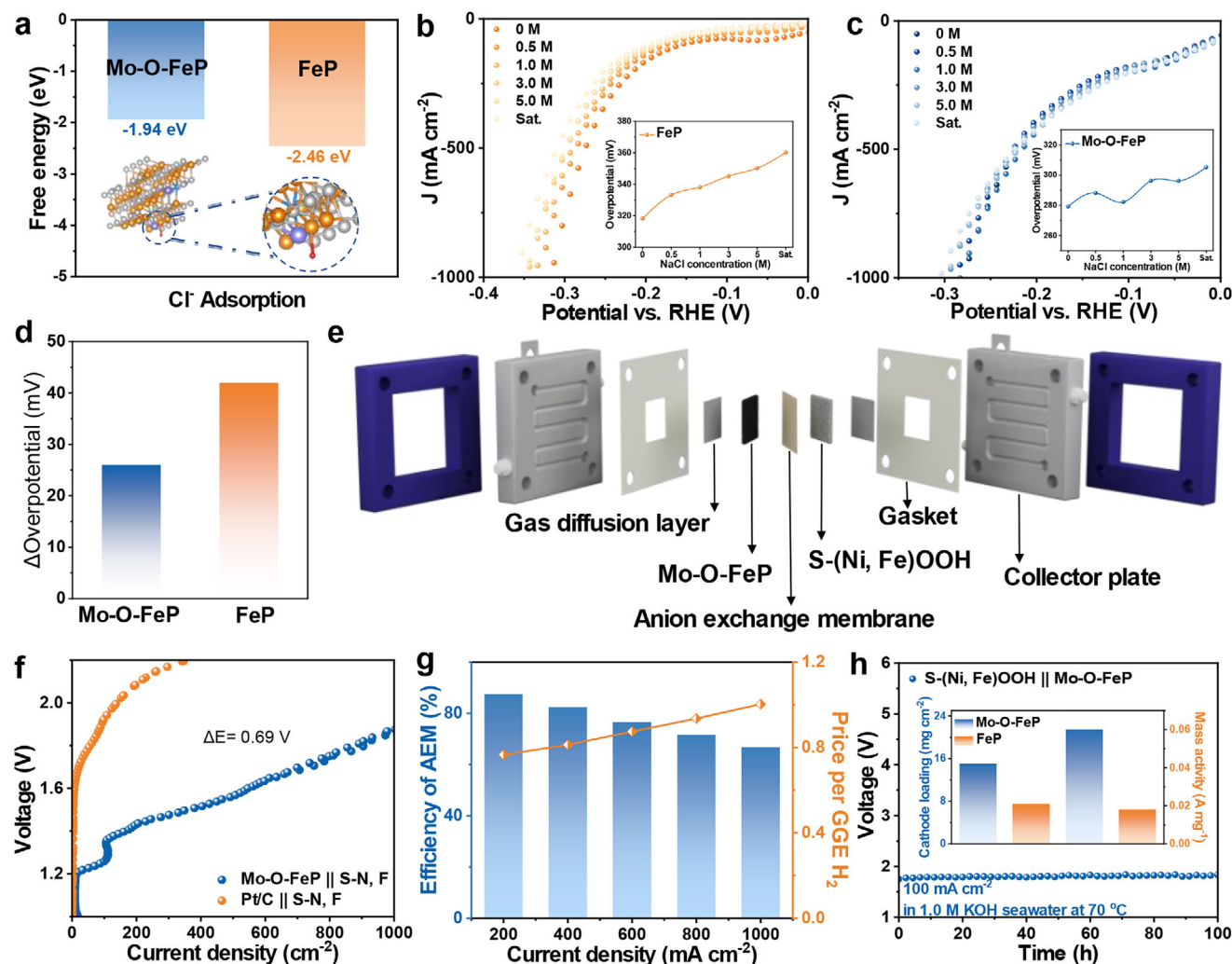


Figure 5. a) Adsorption energy of Cl⁻ at Fe sites on Mo—O—FeP and FeP surfaces. b) FeP and c) Mo—O—FeP for the HER in different Cl⁻ concentrations. d) Corresponding variogram of overpotential at 1000 mA cm⁻². e) Diagram of an AEM electrolyzer. f) Polarization curves of the Mo—O—FeP||S-N, F and Pt/C||S-N, F. g) Efficiency and price of per GGE H₂ of AEM electrolyzer. h) Stability of AEM electrolyzer operating at 100 mA cm⁻².

3. Conclusion

In conclusion, a strategy of Mo—O doping into FeP (Mo—O—FeP) is proposed to change the interfacial water molecular configuration, thus enhancing the kinetics of alkaline HER in seawater. In situ Raman spectroscopy and DFT results show that Mo—O doping changes the configuration of interfacial water, thus promoting water dissociation and hydrogen adsorption. In addition, it also significantly reduces the adsorption energy of Cl⁻ on the Mo—O—FeP catalyst. The Mo—O—FeP only needs the overpotential of 279 mV to achieve the current density of 1000 mA cm⁻², which is much better than FeP and other reported catalysts. When assembled into an AEM electrolyzer, it can drive a current density of 1000 mA cm⁻² at a low cell voltage of 1.87 V, and has good long-term stability at 100 mA cm⁻². This paper provides a novel approach to achieving high activity of HER by changing the configuration of water molecules with element doping.

Supporting Information

Supporting Information is available from the Wiley Online Library or from the author.

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Conflict of Interest

The authors declare no conflict of interest.

Data Availability Statement

The data that support the findings of this study are available from the corresponding author upon reasonable request.

Keywords

asymmetric structure, hydrogen evolution reactions, interfacial water configuration, seawater splitting

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